6.02	**Algebraic formulae**				
6.02a	Formulate algebraic expressions		Formulate simple formulae and expressions from real-world contexts. e.g. Cost of car hire at £50 per day plus 10p per mile. The perimeter of a rectangle when the length is 2 cm more than the width.	*[see, for example, Direct proportion, 5.02a, Inverse proportion, 5.02b, Growth and decay, 5.03a]*	A3, A5, A21, R10
6.02b	Substitute numerical values into formulae and expressions	Substitute positive numbers into simple expressions and formulae to find the value of the subject. e.g. Given that $v = u + at$, find v when $t = 1$, $a = 2$ and $u = 7$.	Substitute positive or negative numbers into more complex formulae, including powers, roots and algebraic fractions. e.g. $v = \sqrt{u^2 + 2as}$ with $u = 2.1$, $s = 0.18$, $a = {}^{-}9.8$.		A2, A5

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…	DfE Ref.
6.02c	Change the subject of a formula	Rearrange formulae to change the subject, where the subject appears once only. e.g. Make d the subject of the formula $c = \pi d$. Make x the subject of the formula $y = 3x - 2$.	Rearrange formulae to change the subject, including cases where the subject appears twice, or where a power or reciprocal of the subject appears. e.g. Make t the subject of the formulae (i) $s = \frac{1}{2}at^2$ (ii) $v = \frac{x}{t}$ (iii) $2ty = t + 1$	*[Examples may include manipulation of algebraic fractions, 6.01g]*	A4, A5
6.02d	Recall and use standard formulae	Recall and use: Circumference of a circle $2\pi r = \pi d$ Area of a circle πr^2	Recall and use: Pythagoras' theorem $a^2 + b^2 = c^2$ Trigonometry formulae $\sin\theta = \frac{o}{h}$, $\cos\theta = \frac{a}{h}$, $\tan\theta = \frac{o}{a}$	Recall and use: The quadratic formula $x = \frac{^{-}b \pm \sqrt{b^2 - 4ac}}{2a}$ Sine rule $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$ Cosine rule $a^2 = b^2 + c^2 - 2bc\cos A$ Area of a triangle 1	A2, A3, A5

GCSE (9–1) content Ref.	Subject content	Initial learning for this qualification will enable learners to…	Foundation tier learners should also be able to…	Higher tier learners should additionally be able to…	DfE Ref.
6.02e	Use kinematics formulae	Use: $v = u + at$ $s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$ where a is constant acceleration, u is initial velocity, v is final velocity, s is displacement from position when $t = 0$ and t is time taken. [Knowledge of the definition of each letter will not be required.]			A2, A3, A5